Simulation benchmark of cue-transfer decoding

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> A freely fit rotation can make transfer decoding (PT) high with nothing shared across cues, but a correspondence-shuffled control rises exactly as much. A shared five-dimensional geometry, matched to the recorded populations in self-decoding and dimensionality, lifts PT 4.3–4.9 times above that control at 50 classes. The recorded V1/V2 populations sit at 1.3–1.5 times.

Simulation design

To ask what successful transfer decoding implies about shared geometry, we ran the unchanged analysis pipeline on synthetic populations whose cross-cue structure is known. Two models bracket the possibilities: nothing shared across cues (Model 1), or a stimulus geometry shared by all cues (Model 3).

Neurons. Each neuron i has a mean spike count r_i and a tuning depth κ_i, the standard deviation of its mean response across stimuli divided by r_i. Both are drawn from log-normal fits to the 415 recorded neurons (330–630 ms window), with κ_i capped at 0.35:

$$r_i = 9.4\, e^{0.66\,\xi_i}, \qquad \kappa_i = 0.14\, e^{0.65\,\eta_i}, \qquad \xi_i,\ \eta_i \sim \mathcal{N}(0,1) \tag{1}$$

Model 1: no shared structure. The mean response of neuron i to stimulus s in cue c is drawn independently for every stimulus and cue. The Gamma distribution keeps rates positive and gives each tuning curve mean r_i and coefficient of variation κ_i:

$$\lambda_{i,s,c} \sim \mathrm{Gamma}\big(\text{shape} = \kappa_i^{-2},\ \text{scale} = r_i \kappa_i^{2}\big) \tag{2}$$

Model 3: shared shape geometry. Each stimulus is a point $\mathbf{z}_s$ in a five-dimensional shape space common to all cues, drawn from a standard normal and then centred and whitened across stimuli. Each neuron reads this point out linearly, along a unit direction drawn independently for each cue:

$$\lambda_{i,s,c} = r_i \left(1 + \kappa_i\, \mathbf{u}_{i,c}^{\top} \mathbf{z}_s\right), \qquad \mathbf{z}_s \in \mathbb{R}^5, \qquad \mathbf{u}_{i,c} \text{ uniform on the unit sphere} \tag{3}$$

Whitening gives each tuning curve mean r_i and standard deviation $r_i\kappa_i$, so every Model 3 neuron matches its Model 1 counterpart. The cues share the geometry of the stimulus points but embed it differently in the population. Where needed, κ_i is lowered further so that no rate is negative.

Trials. Spike counts are Poisson around the mean rate, since the recorded Fano factor was close to 1 (median 1.08):

$$x_{i,s,c,t} \sim \mathrm{Poisson}(\lambda_{i,s,c}), \qquad t = 1, \ldots, 10 \tag{4}$$

Sweep. We varied the number of neurons N (5, 10, 20, ..., 100) and of stimulus classes (5–50), with 16 independent populations per grid point. Five shape dimensions match Model 3 to the recorded populations in dimensionality and self-decoding (Fig. 2); because they need at least six stimuli, Model 3 starts at 10 classes. Each Model 3 population reuses the neurons of the Model 1 population at the same grid point, so the two models differ only in geometry.

Decoding and control. Each cue is z-scored per neuron, and a linear ECOC decoder is trained on the target cue. PT fits a rotation between the centred class means of the source cue (training trials) and the target cue, then decodes one rotated held-out source trial per class. The control applies one random permutation π to both the source class means and the held-out trials before fitting:

$$\hat{R} = \arg\min_{R^\top R = I} \|\mathbf{M}_{\text{tgt}} - \mathbf{M}_{\text{src}} R\|_F$$
$$\hat{R}_\pi = \arg\min_{R^\top R = I} \|\mathbf{M}_{\text{tgt}} - \mathbf{P}_\pi \mathbf{M}_{\text{src}} R\|_F \quad (5)$$

PT decodes a trial of stimulus s rotated by $\hat{R}$ and scores it against label s. The control decodes a trial of stimulus $\pi(s)$ rotated by $\hat{R}_\pi$ and scores it against the same label s. It therefore measures the accuracy a freely fit rotation produces once the stimulus correspondence is destroyed: the overfitting floor.

Without shared structure, PT equals its control

In Model 1 the rotation alone produced substantial transfer, and the control produced exactly as much. At 50 classes (chance 0.02), PT rose from 0.03 at $N = 5$ to 0.30 at $N = 50$ and 0.75 at $N = 100$, while the control reached 0.03, 0.31 and 0.74. Across all 66 grid points PT/control had a median of 1.00, and stayed within 0.95–1.05 for $N \geq 20$ (Fig. 1A, blue).

The control behaved as intended. Scored against each trial's original identity instead of the target label, it stayed within 0.02 of chance, so the permutation did remove the correspondence. Above-chance PT is therefore not, by itself, evidence of shared geometry.

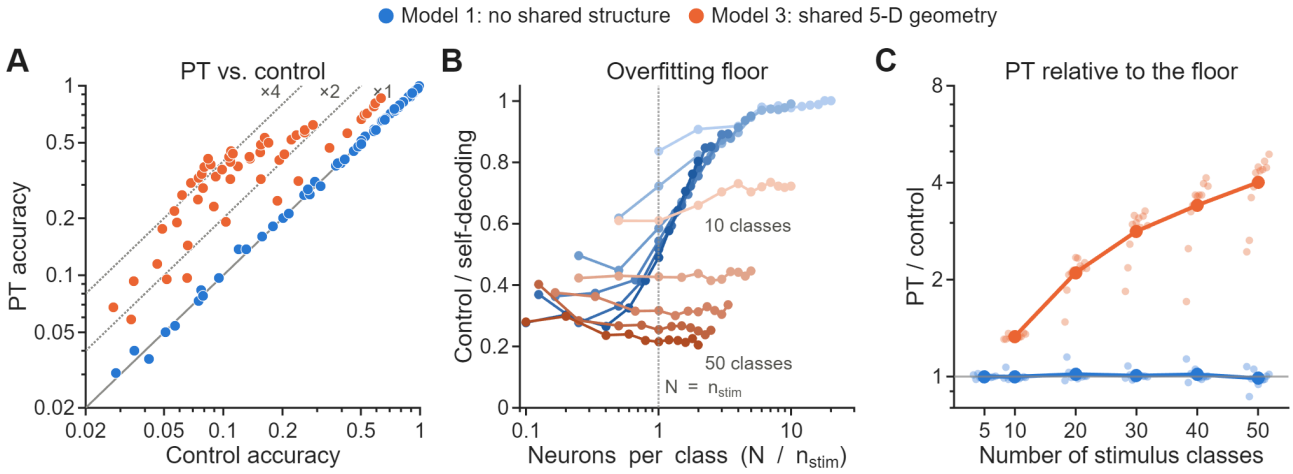


Figure 1. Transfer decoding on synthetic populations with and without shared geometry. Each point averages 16 independent populations and both transfer directions. **A**, PT against the correspondence-shuffled control at every grid point (N = 5–100 neurons; 5–50 classes for Model 1, 10–50 for Model 3); lines mark PT = 1×, 2× and 4× control. **B**, Control accuracy as a fraction of self-decoding against neurons per class; one line per class count, darker lines for more classes. **C**, PT/control against the number of classes; dots are individual N, lines their mean. PT is rotation-only; ratios are ratios of mean accuracies.

Under the null, the floor is set by neurons per class

In Model 1, the control's share of self-decoding grew with the number of neurons per class, N/n_{stim} (Fig. 1B, blue). It was 0.28–0.50 at $N/n_{\text{stim}} \leq 0.25$, 0.49–0.59 at $N = n_{\text{stim}}$ (20–50 classes), 0.75–0.91 at $N = 2n_{\text{stim}}$, and 0.97–1.00 from six neurons per class upward. With 20 or more classes, the curves for different class counts nearly overlapped.

This makes the neuron-to-class ratio quantitative. Once neurons outnumber classes about sixfold, a freely fit rotation recovers essentially all decodable structure although none is shared. The floor also exceeds chance when neurons are fewer than classes (0.04 against 0.02 at $N = 10$ with 50 classes), so $1/n_{\text{stim}}$ is never the right baseline.

Shared geometry lifts PT above the floor

With a shared five-dimensional geometry (Model 3), PT recovered nearly all within-cue decodability from 30 neurons upward (PT/self 0.92–1.00), and 0.62–0.80 with only 5–10 neurons. The control stayed well below PT throughout (Fig. 1A, orange). At 50 classes and $N = 100$, for example, self-decoding was 0.41, PT 0.41 and the control 0.08.

PT/control grew with the number of classes, from 1.33 at 10 classes to 2.10 at 20 and 4.01 at 50 (means over N; Fig. 1C). Beyond 20 neurons it changed little with N (3.59–4.91 at 50 classes). Unlike the null, the floor did not rise with N: with 50 classes, control/self stayed at 0.20–0.30 from 5 to 100 neurons (Fig. 1B, orange).

A plausible reading is that a rotation can partly align a few points whatever their correspondence, but not many points in a low-dimensional configuration. PT/control therefore indexes shared geometry, and its sensitivity grows with the number of classes. A two-dimensional version showed the same pattern but decoded poorly, because its 50 classes crowd into a plane (self-decoding at most 0.17).

Recorded populations fall between the two models

We compared the four recorded populations with both models at 50 classes, using the same decoder and control (Fig. 2, Table 1). In every population PT exceeded the control, by 26–50% at full size, and PT/control was nearly constant across subsampled population sizes (1.15–1.57 from $N = 10$ upward; Fig. 2B). This margin lies between the null (1.00) and the shared-geometry model (3.59–4.91 for $N \geq 20$).

The comparison with Model 3 is at a matched operating point. From 40 to 100 neurons, its self-decoding (0.28–0.41) and participation ratio (7.1–7.8) fall within the recorded ranges (0.26–0.49 and 7.0–9.5; Fig. 2A, C), yet its PT/control is 4.3–4.9. Model 1 reaches the recorded self-decoding at N = 20–40 (0.30–0.52), where its PT/control is 0.98–1.05.

The small recorded margin therefore does not reflect the recordings' signal-to-noise or dimensionality. A fully shared geometry at the same operating point would produce a margin about three times larger.

Table 1. PT and its control at full population size, 50 classes (chance 0.02). Recorded values pool both directions of all three cue pairs; model values average 16 populations at $N = 100$.

Population	N	Self-decoding	PT	Control	PT / control	Participation ratio
FR V1	48	0.26	0.11	0.09	1.31	7.0
FR V2	112	0.34	0.17	0.13	1.29	7.8
KO V1	109	0.49	0.26	0.17	1.50	7.8
KO V2	146	0.37	0.21	0.17	1.26	9.5
Model 1	100	0.92	0.75	0.74	1.02	30.0
Model 3 (5-D)	100	0.41	0.41	0.08	4.91	7.8

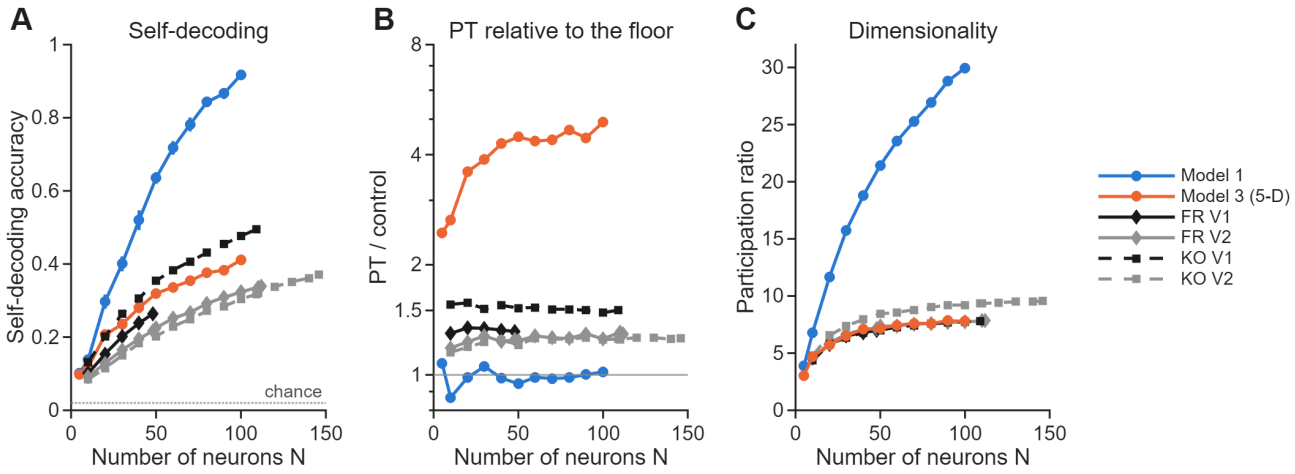


Figure 2. Recorded V1 and V2 populations compared with both models at 50 stimulus classes. Recorded curves subsample neurons from each population (FR solid, KO dashed; V1 black, V2 gray) and use the same decoder and control as the models. Model 3 has a five-dimensional shared latent. **A**, Self-decoding; the dotted line marks chance. **B**, PT/control; the gray line marks PT = control. **C**, Participation ratio of the z-scored, trial-averaged class means. Error bars in A are SEM across synthetic populations (models) or across resamples (recorded data).

Interpretation and limitations

- **PT needs a matched control.** A freely fit rotation recovers most decodable structure once neurons outnumber classes, even when nothing is shared. Above-chance PT alone therefore cannot establish shared geometry.
- **The control is specific.** It matched PT exactly without shared structure and fell well below PT with shared structure, at every N and class count tested.
- **The floor appears to follow the dimensionality of the signal relative to the number of classes, not N itself.** In Model 1 the signal dimensionality grows with N, and so does the floor. In Model 3 it stays at five dimensions, and the floor does not rise. This fits the higher PT/control found when the recorded data are restricted to a few principal components.
- **The recorded populations share part of their geometry.** Their margin over the floor is reliable across population sizes but about three times smaller than that of a fully shared geometry with the same dimensionality and self-decoding. This is consistent with structure that is partly shared and partly rendering-specific.

Limitations. Both models match the recorded single-neuron statistics, and Model 3 also matches their self-decoding and dimensionality, but neither reproduces the recorded margin. A partially shared geometry is the natural next model. Trials were independent Poisson draws, so the simulations contain no noise correlations.